Yilda Boukhtouchen · Ph.D. Candidate in Physics

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Queen's University
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Education

2020 – Present — **Ph.D. Physics, Queen's University** Kingston, ON, Canada.
Advisor: Joseph Bramante, Queen's High Energy Astroparticle Theory Group

2016 – 2020 — **B.Sc. Mathematical Physics, University of Waterloo** Waterloo, ON, Canada.
Minors: Pure Mathematics, Astrophysics.

Research Publications and Preprints

5. Bleau, K., **Boukhtouchen, Y.**, Bramante, J., & Kulkarni, R. (2025). *Secluded dark composites and remnant binding fields*. *Journal of Cosmology and Astroparticle Physics*, *2025*(12), 034. doi:10.1088/1475-7516/2025/12/034

4. **Boukhtouchen, Y.**, Bramante, J., Cappiello, C., & Diamond, M. (2025). *Deconstructive Composite Dark Matter Detection*. arXiv: 2512.16043 [hep-ph]

3. Acevedo, J. F., **Boukhtouchen, Y.**, Bramante, J., Cappiello, C., Mohlabeng, G., & Tyagi, N. (2025). *Loosely bound composite dark matter*. *JCAP*, *03*, 013. doi:10.1088/1475-7516/2025/03/013

2. Acevedo, J. F., An, H., **Boukhtouchen, Y.**, Bramante, J., Richardson, M. L. A., & Sansom, L. (2024). *Dark matter induced baryonic feedback in galaxies*. *Phys. Rev. D*, *110*(8), 083004. doi:10.1103/PhysRevD.110.083004

1. Baum, S., Stengel, P., Abe, N., Acevedo, J. F., Araujo, G. R., Asahara, Y., … Watanabe, H. (2023). *Mineral detection of neutrinos and dark matter, a whitepaper*. *Physics of the Dark Universe*, *41*, 101245. doi:10.1016/j.dark.2023.101245

Presentations

Conference Talks

November 2025 — **TeV Particle Astrophysics (TeVPA)**, IFIC, Valencia, Spain
Having a blast with composites: composite dark matter disassembly in the Earth.

September 2025 — **Invisibles2025 Workshop**, CERN, Geneva, Switzerland
Having a blast with composites: composite dark matter disassembly in the Earth.

July 2025 — **Women in Physics Canada (WIPC+)**, Queen's University, Kingston ON.
Direct detection of composite dark matter.

Dark Side of the Universe (DSU), Université de Montréal, Montréal QC.
Having a blast with composites: composite dark matter disassembly in the Earth.

August 2024 — **TeV Particle Astrophysics (TeVPA)**, University of Chicago, Chicago, IL.
Loosely bound composite dark matter.

Canadian Astroparticle Physics Annual Community Meeting, Arthur B. McDonald Institute, Kingston ON.
New searches for composite dark matter.

May 2023 — **Phenomenology Symposium**, University of Pittsburgh, Pittsburgh, PA.
Dark matter-induced baryonic feedback in galaxies.

Oct 2022 — **Mineral Detection of Dark Matter and Neutrinos Workshop**, IFPU, Trieste, Italy.
Mineral detection of dark matter: feasibility studies at Queen's University.

Public Talks

Jan 2026 — **All about supernovae: the life and death of stars**.
Astronomy on Tap, McDonald Institute, Kingston, ON.

Nov 2024 — **The theory and future of dark matter**.
Dark Matter Day Masterclass, McDonald Institute, Kingston, ON.

Presentations (continued)

Dec 2022 — **Written in the sky? What astronomy tells us about dark matter.**
Observatory Open House, Queen's University, Kingston, ON.

Awards and Achievements

2025	**Harold M. Cave Travel Scholarship**, Queen's University	$1,000
	Harold M. Cave Travel Scholarship, Queen's University	$1,600
2024	**Ontario Graduate Scholarship**, Queen's University	$15,000
	Student Achievement Award: Education and Outreach, McDonald Institute	$250
2023	**Ontario Graduate Scholarship**, Queen's University	$15,000

Teaching Experience

2020 – Present — **Graduate Teaching Assistant** Queen's University.
PHYS 490/891, Nuclear and Particle Physics (Fall 2021, 2022, 2025)
PHYS 345, Quantum Physics (Winter 2023, 2024, 2025)
EMPH 431, Electromagnetic Theory (Fall 2022, 2024)
PHYS 316, Methods in Mathematical Physics (Fall 2021, 2022, 2023)
PHYS 315, Physical Processes in Astrophysics (Winter 2022)
PHYS 435/815, Stellar Structure and Evolution (Fall 2020, 2021)
ASTR 101, Astronomy I: The Solar System (Fall 2020, Winter 2021)

2024 – 2025 — **Lecturer, Summer Particle Astrophysics Workshop** Queen's University.
An Introduction to dark matter. (May 2025)
Direct and indirect dark matter detection. (May 2024)

Community Involvement

2023 – Present — **Director of Operations**, *Innovation, Diversity, Exploration & Advancement in STEM (IDEAS) Initiative* Queen's University.

2025 — **Workshop Organizer**, *Summer Particle Astrophysics Workshop* Queen's University & hybrid.

2023 – Present — **Founding Member and Executive Member**, *Queen's Physics Upper-Year Undergraduate Mentorship Program (Q-PUMP)* Queen's University.

2022 – 2025 — **President**, *Graduate Physics Society* Queen's University.

2021 – 2023 — **Scheduling Coordinator**, *Let's Talk Science* Queen's University.

2022 – 2022 — **Graduate Representative**, *Departmental EDII-F Committee* Queen's University.

2022 — **Conference Volunteer**, *TeV Particle Astrophysics Conference* Queen's University.

2018 – 2020 — **Physics Undergraduate Society (PhysClub) Executive** University of Waterloo.
President, Fall 2019 & Winter 2020 · VP, Winter 2019 · Treasurer, Spring & Fall 2018

Outreach

Feb 2025 — **Organizer and Participant**, *Queen's Physics Research Night*, Queen's University.
Q-PUMP, Queen's Undergraduate Upper-Year Mentorship Program.

July 2023, 2024 — **Camp Director**, *IDEAS Summer Camp*, Queen's University.
IDEAS (Innovation, Diversity, Exploration & Advancement in STEM) Initiative.

Outreach (continued)

Jun 2021 — **Mentor**, *Canadian Online Science Fair* Queen's University.
IDEAS (Innovation, Diversity, Exploration & Advancement in STEM) Initiative.

Feb 2021 — **Materials Assistant**, *Let's Talk Astrophysics Symposium* Queen's University.
Let's Talk Science at Queen's University, and the IDEAS Initiative.

Oct 2020 — **Workshop Creation Volunteer**, *Go ENG Girl* Queen's University.
Let's Talk Science and Queen's Connections Engineering, in partnership with the Ontario Network of Women in Engineering (ONWiE).

2020 – 2021 — **Classroom Volunteer**, *Let's Talk Science*, Queen's University.

2017 - 2019 — **High School Physics Lab Facilitator**, *Science Outreach* University of Waterloo.
Dec 2019, Apr 2018, Apr 2017

2016 - 2019 — **Workshop Volunteer**, *PhysiX: Girls Matter Workshop* University of Waterloo.
Nov 2019, Oct 2018, Dec 2017, Dec 2016

Physics Workshop Facilitator, *Science Open House* University of Waterloo.
Oct 2019, Oct 2018, Oct 2017, Oct 2016